HUNG THAI

AI Engineer

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PROFESSIONAL SUMMARY

AI Engineer with 3 years of experience applying LLMs, OCR and computer vision to production workflows at a US real-estate appraisal management company. Replaced a keyword/regex email classifier with an LLM pipeline using embedding-based question routing, raising automated resolution from 50–60% to 95% across 700–1,300 emails per day. Also delivered document routing over ~10,000 PDFs, a 54-class property image classifier with a monthly retraining loop, and a self-hosted voice agent built solo — owning delivery end to end, including the Ubuntu GPU servers and MariaDB replication the systems run on.

TECHNICAL SKILLS

Languages	Python, SQL, Bash
LLM & NLP	Prompt engineering (chain-of-thought, few-shot), embedding-based retrieval and intent routing, LangChain, LangGraph, Ollama; gpt-oss, Gemma, Qwen, nomic-embed
Vision & OCR	YOLO (detection, segmentation, classification), LayoutLMv3, OpenCV, classical CV (FFT, morphological operations); PaddleOCR, EasyOCR, Tesseract
Speech	FasterWhisper (STT), Kokoro (TTS), Silero VAD, LiveKit, Twilio Media Streams
Infrastructure	Ubuntu server administration, NVIDIA GPU (CUDA, driver setup, fault diagnosis), systemd, Apache, Docker (basic), Git
Databases	MariaDB / MySQL — master–slave replication, dump and restore, replica bootstrap and resync
Frameworks	FastAPI, Selenium, Poetry

PROFESSIONAL EXPERIENCE

AI Engineer — Suntender Valuations, Inc.

2023 – 07/2026 · Remote (US client)

- **Sole engineer for every AI system on the platform** — four running in production (email classification and routing, document classification, property image classification, signature detection) plus a voice-agent prototype — owning each from problem framing through training to day-to-day operation.
- **Translated requirements from non-technical staff into model behaviour.** Worked directly with the client's US-based QC team, who review output daily, to define the labels and decision rules each system should follow, and revised those definitions as edge cases surfaced in real work.
- **Built the labelling and evaluation workflow that keeps the models current**: collecting QC-reported errors as they arise, rebalancing skewed datasets, and running retraining or prompt revision on a regular cycle as new document types and property categories appeared.
- **Delivered every model as a Python service invoked by the company's PHP application** under www-data, handling filesystem permissions, environment isolation, and error/exit-code contracts between the two runtimes; long-running jobs managed with systemd.
- **Owned the supporting infrastructure**: three Ubuntu GPU servers hosting all ML workloads (CUDA and driver setup, monitoring scripts, GPU fault diagnosis), and MariaDB master–slave replication across three replicas (~9–14 GB) including dump/restore bootstrap and resync after replication breaks.

Software Engineer Intern (API Integration) — Mynavi TechTus Vietnam

03/2023 – 06/2023 · Da Nang, Vietnam

- Developed and tested REST APIs for data exchange between internal systems.
- Built Python scripts to call APIs, process responses and automate workflows; handled authentication, error handling and input validation in an Agile/Scrum team.

KEY PROJECTS

Email Classification & Routing Pipeline

Suntender Valuations · in production

Python · gpt-oss:20b via Ollama · LangChain · nomic-embed

- Inherited a keyword/regex classifier that auto-resolved 50–60% of daily email. Rebuilt it as an LLM pipeline that answers ~15 classification questions per email across four recipient roles (appraiser, lender, borrower, point-of-contact), each question carrying 2–4 labels, then merges the answers into a routing decision. **Automated resolution now runs at 95% of 700–1,300 emails per day**, reached over a year of iterating on prompts and retrieval examples against QC feedback.
- **Halved model calls per email with embedding-based question routing.** An offline-built index over labelled example phrasings ranks all 15 questions by relevance to the incoming email, and only the top 7 reach the LLM — removing the throughput bottleneck that appeared as the question set grew.
- **Runs fully on-premise** (temperature 0 for reproducible, testable output) so borrower PII never leaves company infrastructure. Hardened for unattended operation with per-question hard timeouts that fail loudly rather than hang, preprocessing of email signatures, legal disclaimers and marketing footers, and merge safeguards so fee and payment requests are never dropped when answers are combined.

Document Routing & Classification

Suntender Valuations · in production

LayoutLMv3 · PaddleOCR · Python

- Classifies incoming appraisal documents into ~10 report types so they route to the right workflow automatically. Trained on ~10,000 PDFs, reaching **90–98% accuracy per class**.
- Chose a layout-aware model over text-only classification because several report types share nearly the same vocabulary and differ mainly in structure — table position, headers, signature blocks. Rebalanced a heavily skewed dataset by downsampling majority classes and augmenting minority ones with noise, flips and rotation.

Property Image Classification

Suntender Valuations · in production

YOLO (classification) · Python

- Classifies inspection photographs into **54 property-area classes** to verify that each report contains the required photograph of every area an appraiser must inspect, and flags missing or mismatched images for QC. Trained on 500,000+ images, holding **above 90% per class**.
- Built a monthly improvement cycle: QC reports incorrect predictions during their normal review, I relabel those cases and retrain — keeping accuracy stable as new property types appear.

Signature Detection

Suntender Valuations · first production project

OpenCV · FFT · OCR · later YOLO

- Detects signature blocks to validate that appraisal reports are complete. Built before object-detection tooling was available to me: removed ruled lines morphologically, suppressed printed text using OCR bounding boxes, then used FFT analysis to locate the remaining large irregular blob regions. Later rebuilt with YOLO object detection once the tooling matured.

Real-Time Voice & Text AI Agent

Sole developer · working prototype

Python · LangGraph · LiveKit · FasterWhisper · Kokoro · Silero VAD · FastAPI · Twilio

- Built alone: a self-hosted voice and text assistant for appraisal customers, running entirely on one GPU server with no cloud AI services. One LangGraph state machine handles identification, order lookup and task routing for three caller roles and is shared by both channels, so the conversation logic exists once.
- Embedding-based intent routing answers the request without invoking the LLM when the top match is confidently ahead of the rest, and falls back to the LLM only when ambiguous. Cut perceived call latency further by pre-rendering fixed replies to cached audio and playing short fillers during slow database lookups.
- Also places outbound appointment-confirmation and point-of-contact calls, including LLM-based voicemail-versus-human detection, and bridges real inbound phone calls into the same conversation engine through Twilio Media Streams.

EDUCATION & LANGUAGES

Bachelor of Engineering, Software Engineering — Da Nang University of Science and Technology, graduated 2026

Vietnamese native · **English** professional working proficiency (TOEIC 610), used daily with a US-based client since 2023